
```

clc; clear; close all;

% Part 1a

x = 5;
n_terms = 10;

term_idx = (1:n_terms)';
approx_val = zeros(n_terms, 1);
true_rel_err = zeros(n_terms, 1);
approx_rel_err = zeros(n_terms, 1);

true_val = exp(x);
current_sum = 0;

for i = 1:n_terms
    k = i - 1;
    term = (x^k) / factorial(k);
    current_sum = current_sum + term;
    approx_val(i) = current_sum;

    % True relative error: err = |(e_n - e^x) / e^x|
    true_rel_err(i) = abs((current_sum - true_val) / true_val);

    % Approximate relative error: err_rel = |(e_{n+1} - e_n) / e_{n+1}|
    if i > 1
        approx_rel_err(i) = abs((approx_val(i) - approx_val(i-1)) /
approx_val(i));
    else
        approx_rel_err(i) = NaN; % No previous term for the first entry
    end
end

ErrorTable = table(term_idx, approx_val, true_rel_err, approx_rel_err, ...
    'VariableNames', {'Num_Terms_n', 'Approximation', 'True_Rel_Err',
    'Approx_Rel_Err'});

disp('Maclaurin Series Approximation Table:');
disp(ErrorTable);

```

Maclaurin Series Approximation Table:

Num_Terms_n	Approximation	True_Rel_Err	Approx_Rel_Err
1	1	0.99326	NaN
2	6	0.95957	0.83333
3	18.5	0.87535	0.67568
4	39.333	0.73497	0.52966
5	65.375	0.55951	0.39834
6	91.417	0.38404	0.28487
7	113.12	0.23782	0.19185
8	128.62	0.13337	0.12052
9	138.31	0.068094	0.070048

10

143.69

0.031828

0.037458

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